1.1

Output

```
[1] 328
[1] "priya"
[1] 8.02
[1] TRUE

=== Code Execution Successful ===
```

1.2

Output

```
[1] 4398
[1] 430000.8
[1] "IT"
[1] TRUE

=== Code Execution Successful ===
```

1.3

Output

```
[1] "Moblie"
[1] 35000
[1] 25
[1] FALSE

=== Code Execution Successful ===
```

1.4

Output

```
[1] "Tirupati"
[1] 38.5
[1] FALSE
[1] 10.4

=== Code Execution Successful ===
```

1.5

Output

```
[1] "Science"
[1] 100
[1] 88
[1] TRUE

=== Code Execution Successful ===
```

1.6

Output

```
[1] 435022348
[1] "Rahul"
[1] 32000.79
[1] TRUE

=== Code Execution Successful ===
```

1.7

Output

```
[1] "Interseller"
[1] 9.2
[1] 2700
[1] TRUE

=== Code Execution Successful ===
```

1.8

Output

```
[1] 280
[1] "Highway Road"
[1] 50
[1] TRUE

=== Code Execution Successful ===
```

2.1

Output

```
[1] "Eligible for Loan"

=== Code Execution Successful ===
```

2.2

Output

```
[1] "Not Eligible for Exam"

=== Code Execution Successful ===
```

2.3

Output

```
[1] "Discount Applied"

=== Code Execution Successful ===
```

2.4

Output

```
[1] "Medium Usage"

=== Code Execution Successful ===
```

2.5

Output

```
[1] "Bonus Eligible"

=== Code Execution Successful ===
```

2.6

Output

```
[1] "Valid Password"

=== Code Execution Successful ===
```

2.7

Output

```
[1] "Extreme Temperature Warning"

=== Code Execution Successful ===
```

2.8

Output

```
[1] "B"

=== Code Execution Successful ===
```

2.9

Output

```
[1] "Credit Card Approved"

=== Code Execution Successful ===
```

2.10

Output

```
[1] "Low Risk"

=== Code Execution Successful ===
```

2.11

Output

```
[1] "Medium Green Signal"

=== Code Execution Successful ===
```

2.12

Output

```
[1] "Not Eligible for Appraisal"

=== Code Execution Successful ===
```

2.13

Output

```
[1] "Exam Allowed"

=== Code Execution Successful ===
```

2.14

Output

```
[1] "Surge Pricing Applied"

=== Code Execution Successful ===
```

2.15

Output

```
[1] "Machine Shutdown"

=== Code Execution Successful ===
```

2.16

Output

```
[1] "Admission Granted"

=== Code Execution Successful ===
```

3.1

Output

```
[1] 1000

=== Code Execution Successful ===
```

3.2

Output

```
[1] "Pass"

=== Code Execution Successful ===
```

3.3

Output

```
[1] 22500

=== Code Execution Successful ===
```

3.4

Output

```
[1] 1750

=== Code Execution Successful ===
```

3.5

Output

```
[1] 4410

=== Code Execution Successful ===
```

3.6

Output

```
[1] 102.2

=== Code Execution Successful ===
```

3.7

Output

```
[1] "B"

=== Code Execution Successful ===
```

3.8

Output

```
[1] "Eligible to Vote"

=== Code Execution Successful ===
```

3.9

Output

```
[1] 7687.11

=== Code Execution Successful ===
```

3.10

Output

```
[1] 9922.5

=== Code Execution Successful ===
```

3.11

Output

```
[1] 3700

=== Code Execution Successful ===
```

3.12

Output

```
[1] "First Class"

=== Code Execution Successful ===
```

3.13

Output

```
[1] 13695

=== Code Execution Successful ===
```

3.14

Output

```
[1] 5200

=== Code Execution Successful ===
```

3.15

Output

```
[1] 62920

=== Code Execution Successful ===
```

3.16

Output

```
[1] "Critical"

=== Code Execution Successful ===
```

4.1

Output

```
  RollNo        Name Dept Marks
1    201       Rahul  CSE    65
2    202       Meera  ECE    95
3    203 Karthikaya   IT    90
  RollNo        Name Dept Marks
1    201       Rahul  CSE    65
2    202       Meera  ECE    95
3    203 Karthikaya   IT    90
4    504     Anitha  CSE    88

=== Code Execution Successful ===
```

4.2

Output

```
  EmpID  Name    Dept Salary
1   121  Amit      HR  39000
2   221 Sneha      IT  55000
3   321   Raj Finance  60000
  EmpID  Name    Dept Salary
1   121  Amit      HR  39000
2   221 Sneha      IT  55000
3   321   Raj Finance  60000
4     4  Neha      IT  48000

=== Code Execution Successful ===
```

4.3

Output

```
  PatientID    Name Age  Cost
1       201  Suresh  45 12000
2       202 Lakshmi  38 15000
  PatientID    Name Age  Cost
1       201  Suresh  45 12000
2       202 Lakshmi  38 15000
3       203   Arjun  50 18000

=== Code Execution Successful ===
```

4.4

Output

```
  ProductID ProductName Quantity
1         1      Pencil      100
2         2        Book       50
3         3         Bag       30
  ProductID ProductName Quantity
1         1      Pencil      100
2         2        Book       50
3         3         Bag       30
4         4      Bottle       60

=== Code Execution Successful ===
```

4.5

Output

```
  RollNo     Name Dept Sem1 Sem2 Attendance
1      1    Arjun  CSE   79   80         92
2      2     Sita  ECE   88   85         88
3      3 Charitha   IT   90   88         95
  RollNo     Name Dept Sem1 Sem2 Attendance
1      1    Arjun  CSE   79   80         92
2      2     Sita  ECE   88   85         88
3      3 Charitha   IT   90   88         95
4      4  Deepika  CSE   85   87         90
5      5   Kishan   IT   89   91         93

=== Code Execution Successful ===
```

4.6

Output

```
  EmpID     Name Dept Basic  HRA   DA Rating
1   101   Charan   IT 30000 5000 3000    4.5
2   102 Snehitha   HR 35000 6000 3400    4.2
  EmpID     Name    Dept Basic  HRA   DA Rating
1   101   Charan      IT 30000 5000 3000    4.5
2   102 Snehitha      HR 35000 6000 3400    4.2
3   103 Shashank Finance 49000 9000 4000    4.8

=== Code Execution Successful ===
```

4.7

Output

```
  PatientID    Name Age      Disease Amount
1      2071  Suresh  49     Diabetes  15000
2      2872 Lakshmi  34 Hypertension  12000
  PatientID    Name Age      Disease Amount
1      2071  Suresh  49     Diabetes  15000
2      2872 Lakshmi  34 Hypertension  12000

=== Code Execution Successful ===
```

4.8

Output

```
  ProductID ProductName UnitsSold Revenue
1         1      Mobile       120 1200000
2         2      Laptop        80 4000000
3         3      Tablet        60 1800000
  ProductID ProductName UnitsSold Revenue
1         1      Mobile       120 1200000
2         2      Laptop        80 4000000
3         3      Tablet        60 1800000
4         4 Smart Watch        90  900000

=== Code Execution Successful ===
```

4.9

Output

```
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       45   32           4.5
2       2 Wheat       38   29           3.8
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       45   32           4.5
2       2 Wheat       38   29           3.8
3       3 Maize       40   31           4.1

=== Code Execution Successful ===
```

4.10

Output

```
  RollNo    Name Dept Sem1 Sem2 Attendance
1      1    Arul  CSE   78   80         92
2      2 Bharath  ECE   82   85         88
3      3   Karen   IT   90   88         95
ERROR!
Error: unexpected input in "✔"
Execution halted

=== Code Exited With Errors ===
```

4.11

Output

```
  EmpID  Name    Dept Basic  HRA   DA Rating
1   101 Rahul      IT 30000 6000 3000    4.5
2   102 Sneha      HR 35000 7000 3500     NA
3   103 Kiran Finance    NA   NA   NA    4.8

=== Code Execution Successful ===
```

4.12

Output

```
  PatientID    Name Age  Disease Amount
1       201  Ramesh  45 Diabetes  15000
2       202 Lakshmi  38     <NA>  12000
3       203   Arjun  52  Cardiac  25000

=== Code Execution Successful ===
```

4.13

Output

```
  ProductID ProductName UnitsSold Revenue
1         1      Mobile       120 1200000
2         2      Laptop        80 4000000
3         3      Tablet        60      NA
4         4 Smart Watch        NA  900000

=== Code Execution Successful ===
```

4.14

Output

```
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       45   32           4.5
2       2 Wheat       38   29            NA
3       3 Maize       NA   NA           4.1

=== Code Execution Successful ===
```

5.1

Output

```
[1] "RAM"    "BALA"   "CHITRA"
[1] 3 4 6

=== Code Execution Successful ===
```

5.2

Output

```
[1] "arul"  "meera" "rahul"

=== Code Execution Successful ===
```

5.3

Output

```
[1] "R Programming" "Data Science"  "AI"

=== Code Execution Successful ===
```

5.4

Output

```
[1]  TRUE  TRUE FALSE

=== Code Execution Successful ===
```

5.5

Output

```
A module you have imported isn't available at the moment. It will be available soon.
```

5.6

Output

```
[1] "Strong Password"

=== Code Execution Successful ===
```

5.7

Output

```
[1] 6

=== Code Execution Successful ===
```

5.8

Output

```
[1] "csv" "pdf" "png"

=== Code Execution Successful ===
```

5.9

Output

```
[1] "Gou" "Kar" "Anu"

=== Code Execution Successful ===
```

5.10

Output

```
[1] "Bala" "Ravi"

=== Code Execution Successful ===
```

5.11

Output

```
        Name            Email Department
1  Gayathri    Gayu@UNIV.edu        cse
2      RAVI    ravi@univ.edu       ECE
3      meena meena@gmail.com       Cse

=== Code Execution Successful ===
```

5.12

Output

```
       Product        Code ProductCode
1 Lenovo Phone PRD-101-APL         APL
2   Samsung-Tv PRD-202-SAM         SAM
3  Dell Laptop PRD-303-DEL         DEL

=== Code Execution Successful ===
```

5.13

Output

```
  User              Comment                  CleanComment
1    A     this app is bad     this app is unacceptable
2    B very bad experience very unacceptable experience
3    C      not bad at all      not unacceptable at all

=== Code Execution Successful ===
```

6.1

Output

```
[1] 396
[1] 79.2
[1] 92

=== Code Execution Successful ===
```

6.2

Output

```
[1] 32
[1] 37

=== Code Execution Successful ===
```

6.3

Output

```
$ID
[1] 891

$Name
[1] "Goutham"

$Salary
[1] 49000

$Department
[1] "IT"

[1] "Goutham"
[1] 49000

=== Code Execution Successful ===
```

6.4

Output

```
$PatientID
[1] 6782

$Name
[1] "Ram"

$Age
[1] 47

$TestResults
[1] 149 160 130

[1] 149 160 130

=== Code Execution Successful ===
```

6.5

Output

```
  RollNo    Name Dept Marks
1      1  Anjali  CSE    85
2      2 Koushik  ECE    78
3      3 Manisha   IT    92

=== Code Execution Successful ===
```

6.6

Output

```
[1] 170000

=== Code Execution Successful ===
```

6.7

Output

```
[1] 2200 1050  760
$CustomerID
[1] 38901

$Name
[1] "Anjali"

$City
[1] "Chennai"

     Item Price
1   Phone  2200
2 Headset  1050
3 Charger   760

=== Code Execution Successful ===
```

6.8

Output

```
$Name
[1] "Arun"

$RollNo
[1] 105

$Marks
[1] 78 85 88

  RollNo Total
1    101   240
2    102   255
3    103   270

=== Code Execution Successful ===
```

6.9

Output

```
[1] 5000 5200 5100
$AccountNo
[1] 12345

$Name
[1] "Suresh"

   Date Amount
1 01-01   -500
2 02-01   1000
3 03-01   -200

=== Code Execution Successful ===
```

6.10

Output

```
[1] 85 90 88
$Name
[1] "Gayathri"

$Age
[1] 21

$Scores
[1] 85 90 88

  Subject Marks
1    Math    85
2      CS    90
3 Physics    88

=== Code Execution Successful ===
```

```
> write.csv(sales, "sales_data.csv", row.names = FALSE)
> # Read the CSV file
> sales <- read.csv("sales_data.csv")
> # View first few rows
> head(sales)
        Date Product    Category
1 2026-01-01   Phone Electronics
2 2026-01-01     Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03     Bag Accessories
6 2026-01-03   Phone Electronics
  Quantity Price
1        5 20000
2       10  1500
3        3 50000
4        7  2500
5        5  1500
6        3 20000
> # Check structure and summary
> str(sales)
'data.frame':	6 obs. of  5 variables:
 $ Date    : chr  "2026-01-01" "2026-01-01" "2026-01-02" "2026-01-02"
...
 $ Product : chr  "Phone" "Bag" "Laptop" "Headset" ...
 $ Category: chr  "Electronics" "Accessories" "Electronics" "Electronic
s" ...
 $ Quantity: int  5 10 3 7 5 3
 $ Price   : int  20000 1500 50000 2500 1500 20000
> summary(sales)

> summary(sales)
     Date
 Length:6
 Class :character
 Mode  :character


   Product
 Length:6
 Class :character
 Mode  :character


   Category
 Length:6
 Class :character
 Mode  :character


    Quantity         Price
 Min.   : 3.0   Min.   : 1500
 1st Qu.: 3.5   1st Qu.: 1750
 Median : 5.0   Median :11250
 Mean   : 5.5   Mean   :15917
 3rd Qu.: 6.5   3rd Qu.:20000
 Max.   :10.0   Max.   :50000
> # View cleaned data
```

```
> head(sales)
        Date Product    Category
1 2026-01-01   Phone Electronics
2 2026-01-01     Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03     Bag Accessories
6 2026-01-03   Phone Electronics
  Quantity Price
1        5 20000
2       10  1500
3        3 50000
4        7  2500
5        5  1500
6        3 20000
> # Remove extra spaces in Product and Category columns
> sales$Product <- trimws(sales$Product)
> sales$Category <- trimws(sales$Category)
> # Convert Date column to Date format
> sales$Date <- as.Date(sales$Date, format="%Y-%m-%d")
> # Add a column for total sales per row
> sales$TotalSales <- sales$Quantity * sales$Price
> # View cleaned data
> head(sales)
        Date Product    Category
1 2026-01-01   Phone Electronics
2 2026-01-01     Bag Accessories
3 2026-01-02  Laptop Electronics
4 2026-01-02 Headset Electronics
5 2026-01-03     Bag Accessories
```

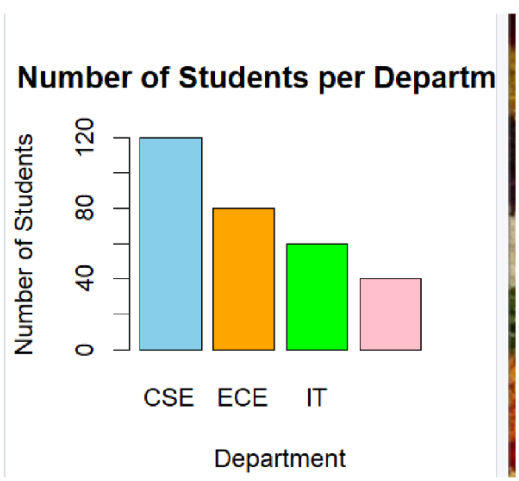
```
5 2026-01-03     Bag Accessories
6 2026-01-03   Phone Electronics
  Quantity Price TotalSales
1        5 20000     100000
2       10  1500      15000
3        3 50000     150000
4        7  2500      17500
5        5  1500       7500
6        3 20000      60000
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, su
m)
> total_sales_product
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
```

```
> # Calculate total sales for each product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, su
m)
> total_sales_product
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
>  Product with Highest Sales
```

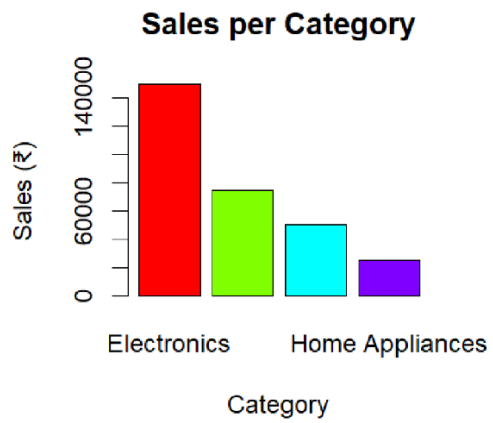
```
> top_product <- total_sales_product[which.max(total_sales_product$Tota
lSales), ]
> top_product
  Product TotalSales
4   Phone     160000
```

```
> quantity_per_category <- aggregate(Quantity ~ Category, data=sales, s
um)
> quantity_per_category
     Category Quantity
1 Accessories       15
2 Electronics       18
> high_sales <- total_sales_product[total_sales_product$TotalSales > 10
000, ]
> high_sales
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
>
```

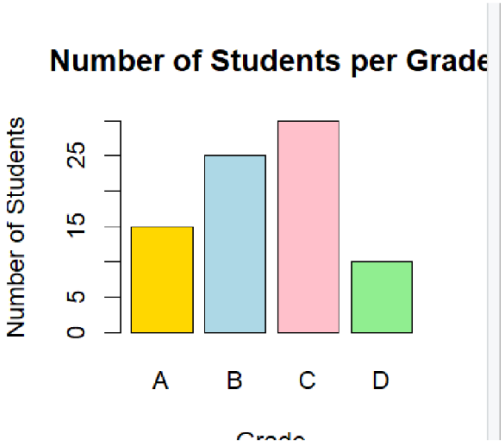
8.1:



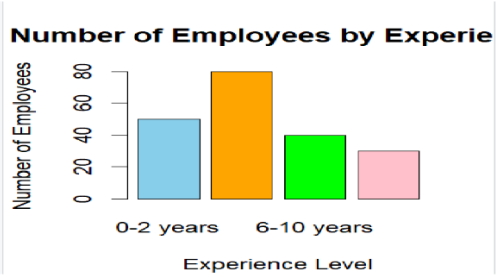
8.2:


Sales per Category
Sales (₹)
0
60000
140000
Electronics
Home Appliances
Category

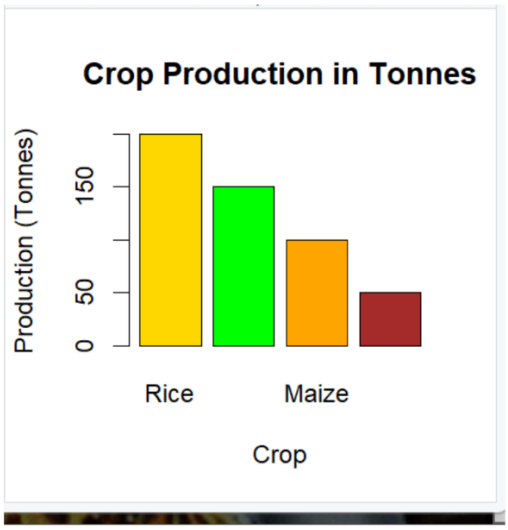
8.3:


Number of Students per Grade
Number of Students
0
5
15
25
A
B
C
D

8.4:


Number of Employees by Experie
Number of Employees
0
20
40
60
80
0-2 years
6-10 years
Experience Level

8.5:


Crop Production in Tonnes
Production (Tonnes)
0
50
150
Rice
Maize
Crop

9.1

Output

```
  Maths Science English
   83.9    80.0    79.5
  Maths Science English
   85.5    80.5    84.0
    Maths   Science   English
 8.265726  9.820613 12.240643
    Name Maths Science English Total
4 Anitha    95      92      92   279
       Name Maths Science English Total
2     Meena    92      80      70   242
4    Anitha    95      92      92   279
5     Kiran    85      89      89   263
6     Sneha    88      82      84   254
8  Gayathri    92      94      61   247
10  Lakshmi    86      81      84   251

=== Code Execution Successful ===
```

9.2

Output

```
   Product Mean Max Min Variance
1    Phone  5.8  10   1     10.7
2   Laptop  3.8   6   2      2.7
3  Headset  6.4   8   5      1.8
4      Bag 10.0  16   2     26.5
5   Camera  3.8  11   1     16.7
6    Mouse  6.0   8   4      2.5
7 Keyboard  7.0   8   6      1.0
  Product Mean Max Min Variance
4     Bag   10  16   2     26.5

=== Code Execution Successful ===
```

9.3

Output

```
[1] 45.4
[1] 46.5
[1] 29 60
[1] 9.547542
  PatientID   Name Age
3       203 Aameer  52
5       205  Ragul  60
7       207  Naren  55

=== Code Execution Successful ===
```

9.4

Output

```
     Dept Salary.Mean Salary.Median
1 Finance    42333.33         42000
2      HR    36000.00         36000
3      IT    31500.00         31500
  EmpID     Name    Dept Salary
3   103    Kiran Finance  40000
6   106 Gayathri Finance  42000
8   108  Lakshmi      HR  37000
9   109    Arjun Finance  45000

=== Code Execution Successful ===
```

9.5

Output

```
[1] 4.1
[1] 4
[1] 4
[1] 0.5444444
  ProductID Product Rating
2         2  Laptop      5
5         5  Camera      5
8         8 Charger      5

=== Code Execution Successful ===
```